Evaluation runs: replog-co (replicated-log-esop20)

generated 2026-11-03 from brain-data 4f70fdcf4 2026-11-03 22:15:02 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{spec}, G_{impl} = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G^{expl}_{spec}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
1	4	8	conforms	0.0	4	1 h, 250 GB	1	1	brain03, 2026-11-03
2	16	128	conforms	0.0	2	1 h, 250 GB	4	28	brain03, 2026-11-03
3	144	4608	conforms	0.6	4	1 h, 250 GB	36	2556	brain03, 2026-11-03
4	2304	294912	conforms	48.1	24	1 h, 250 GB	577	663000	brain03, 2026-11-03
5	57600	29491200	conforms	17518.0	118	48 h, 250 GB	14905	423756076	brain04, 2026-11-01

stateless: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
1	1	8	conforms	0.0	4	1 h, 250 GB	1	1	brain03, 2026-11-03
2	4	128	conforms	0.0	4	1 h, 250 GB	4	10	brain03, 2026-11-03
3	36	4608	conforms	0.6	4	1 h, 250 GB	36	666	brain03, 2026-11-03
4	576	294912	conforms	26.0	25	1 h, 250 GB	576	166176	brain03, 2026-11-03
5	14400		🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-03

stateless: pair **early-reply**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
1	4	32	24	violation	0.0	4	1 h, 250 GB	25	97	brain03, 2026-11-03
2	16	512	384	violation	0.4	4	1 h, 250 GB	388	6172	brain03, 2026-11-03
3	144	18432	13824	violation	70.9	17	1 h, 250 GB	13860	1993212	brain03, 2026-11-03
4	2304			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-03

stateless: pair **early-reply**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
1	1	32	24	violation	0.0	4	1 h, 250 GB	25	25	brain03, 2026-11-03
2	4	512	384	violation	0.2	6	1 h, 250 GB	388	1546	brain03, 2026-11-03
3	36	18432	13824	violation	27.0	16	1 h, 250 GB	13860	498330	brain03, 2026-11-03
4	576			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-03

Not run (stateless). skipped: a smaller size had reached its limit: 5 early-reply fifo, 5 early-reply bag, 6 correct fifo, 6 early-reply fifo, 6 early-reply bag, 6 correct bag, 7 correct bag, 7 correct fifo, 7 early-reply fifo, 7 early-reply bag, 8 correct fifo, 8 correct bag, 8 early-reply bag, 8 early-reply fifo, 9 correct bag, 9 early-reply bag, 9 correct fifo, 9 early-reply fifo.

planned but never scheduled (grid points beyond the ladder): 5 early-reply bag, 5 correct bag, 6 early-reply bag, 6 correct bag, 7 correct bag.

stateful

stateful: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
1	4	8	conforms	0.6	2	1 h, 250 GB	brain02, 2026-11-03
2	16	128	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-03
3	144	4608	conforms	0.6	2	1 h, 250 GB	brain02, 2026-11-03
4	2304	294912	conforms	21.2	26	1 h, 250 GB	brain02, 2026-11-03
5	57600	29491200	conforms	599.7	122	48 h, 250 GB	brain05, 2026-11-02
6			(1 h)			1 h, 250 GB	brain02, 2026-11-03

stateful: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
1	1	8	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-03
2	4	128	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-03
3	36	4608	conforms	0.6	2	1 h, 250 GB	brain02, 2026-11-03
4	576	294912	conforms	20.7	27	1 h, 250 GB	brain02, 2026-11-03
5	14400	29491200	conforms	1049.8	116	1 h, 250 GB	brain02, 2026-11-03
6			(1 h)			1 h, 250 GB	brain02, 2026-11-03

stateful: pair **early-reply**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
1	4	32	24	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
2	16	512	384	violation	0.1	2	1 h, 250 GB	brain02, 2026-11-03
3	144	18432	13824	violation	1.9	2	1 h, 250 GB	brain02, 2026-11-03
4	2304	1179648	884736	violation	49.4	30	1 h, 250 GB	brain02, 2026-11-03
5	57600	117964800	88473600	violation	2064.1	621	48 h, 250 GB	brain05, 2026-11-02
6				(48 h)			48 h, 250 GB	brain04, 2026-11-03

stateful: pair **early-reply**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
1	1	32	24	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
2	4	512	384	violation	0.1	2	1 h, 250 GB	brain02, 2026-11-03
3	36	18432	13824	violation	2.0	6	1 h, 250 GB	brain02, 2026-11-03
4	576	1179648	884736	violation	49.8	30	1 h, 250 GB	brain02, 2026-11-03
5	14400	117964800	88473600	violation	3571.1	619	1 h, 250 GB	brain02, 2026-11-03
6				(48 h)			48 h, 250 GB	brain04, 2026-11-03

Not run (stateful). left out by the table rules (ok): 5 correct bag, 5 early-reply bag, 6 correct bag.

skipped: a smaller size had reached its limit: 7 correct bag, 7 correct fifo, 7 early-reply fifo, 7 early-reply bag, 8 correct fifo, 8 correct bag, 8 early-reply bag, 8 early-reply fifo, 9 correct bag, 9 early-reply bag, 9 early-reply fifo, 9 correct fifo.

the tool ended with an error (exit-101): 7 correct bag, 7 early-reply bag, 8 correct bag, 8 early-reply bag, 9 correct bag.

left out by the table rules (timeout): 6 early-reply bag.

offline

offline: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	4	8	conforms	0.3	4	10 min, 16 GB	0.3	0.0	0.0	in-memory	brain05, 2026-11-02
2	16	128	conforms	0.2	4	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
3	144	4608	conforms	0.7	4	10 min, 16 GB	0.1	0.7	0.0	in-memory	brain05, 2026-11-02
4	2304	294912	conforms	18.1	6	1 h, 250 GB	0.5	18.0	0.1	merge	brain05, 2026-11-03
5	57600	29491200	conforms	713.0	136	1 h, 250 GB	5.1	711.5	1.5	merge	brain05, 2026-11-03

offline: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	1	8	conforms	0.2	4	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain05, 2026-11-02
2	4	128	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
3	36	4608	conforms	0.7	4	1 h, 250 GB	0.1	0.7	0.0	merge	brain05, 2026-11-03
4	576	294912	conforms	17.0	4	1 h, 250 GB	0.2	16.9	0.1	merge	brain05, 2026-11-03
5	14400	29491200	conforms	736.4	136	1 h, 250 GB	1.8	734.9	1.5	merge	brain05, 2026-11-03

offline: pair **early-reply**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	4	32	3	violation	0.3	4	10 min, 16 GB	0.3	0.1	0.0	in-memory	brain05, 2026-11-02
2	16	512	12	violation	0.2	4	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
3	144	18432	108	violation	1.8	4	1 h, 250 GB	0.1	1.8	0.0	merge	brain05, 2026-11-03
4	2304	1179648	1728	violation	41.5	10	1 h, 250 GB	0.5	41.4	0.1	merge	brain05, 2026-11-03
5	57600	117964800	43200	violation	2073.0	572	1 h, 250 GB	5.1	2069.4	3.6	merge	brain05, 2026-11-03

offline: pair **early-reply**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	1	32	3	violation	0.1	4	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain05, 2026-11-02
2	4	512	12	violation	0.3	4	10 min, 16 GB	0.1	0.3	0.0	in-memory	brain05, 2026-11-02
3	36	18432	108	violation	1.8	4	1 h, 250 GB	0.1	1.8	0.0	merge	brain05, 2026-11-03
4	576	1179648	1728	violation	42.1	14	1 h, 250 GB	0.2	42.0	0.1	merge	brain05, 2026-11-03
5	14400	117964800	43200	violation	1960.9	567	1 h, 250 GB	1.8	1957.3	3.6	merge	brain05, 2026-11-03

Not run (offline). planned but never scheduled (grid points beyond the ladder): 5 early-reply bag, 5 correct bag, 6 early-reply fifo, 6 early-reply bag, 6 correct bag, 7 correct bag.

skipped: a smaller size had reached its limit: 6 correct fifo, 6 early-reply fifo, 6 early-reply bag, 6 correct bag, 7 correct bag, 7 correct fifo, 7 early-reply fifo, 7 early-reply bag, 8 correct fifo, 8 correct bag, 8 early-reply bag, 8 early-reply fifo, 9 correct bag, 9 early-reply bag, 9 correct fifo, 9 early-reply fifo.

cross offline, Java implementation against the Rust specification

cross offline, Java implementation against the Rust specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	4	8	conforms	3.4	185	10 min, 16 GB	0.3	3.3	0.0	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	1	8	conforms	3.4	168	10 min, 16 GB	0.1	3.3	0.0	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **early-reply**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	4	32	3	violation	3.4	154	10 min, 16 GB	0.3	3.3	0.0	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **early-reply**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	1	32	3	violation	3.4	166	10 min, 16 GB	0.1	3.4	0.0	in-memory	brain01, 2026-11-02

Not run (cross offline, Java implementation against the Rust specification). refused (an input missing at run time): 2 correct bag, 2 early-reply bag, 2 early-reply fifo, 2 correct fifo, 3 correct bag, 3 early-reply bag, 3 correct fifo, 3 early-reply fifo.

planned but never scheduled (grid points beyond the ladder): 5 early-reply bag, 5 correct bag, 6 early-reply bag, 6 correct bag, 7 correct bag.

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	4	8	conforms	3.1	165	10 min, 16 GB	3.1	0.0	0.0	in-memory	brain01, 2026-11-02
2	16	128	conforms	3.4	166	10 min, 16 GB	3.4	0.1	0.0	in-memory	brain01, 2026-11-02
3	144	4608	conforms	3.8	166	10 min, 16 GB	3.8	0.7	0.0	in-memory	brain01, 2026-11-02

cross offline, Rust implementation against the Java specification: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	1	8	conforms	0.7	160	10 min, 16 GB	0.7	0.2	0.0	in-memory	brain01, 2026-11-02
2	4	128	conforms	3.1	160	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-02
3	36	4608	conforms	3.9	158	10 min, 16 GB	3.9	0.7	0.0	in-memory	brain01, 2026-11-02

cross offline, Rust implementation against the Java specification: pair **early-reply**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	4	32	3	violation	3.1	165	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-02
2	16	512	12	violation	3.4	166	10 min, 16 GB	3.4	0.1	0.0	in-memory	brain01, 2026-11-02
3	144	18432	108	violation	3.8	166	10 min, 16 GB	3.8	1.8	0.0	in-memory	brain01, 2026-11-02

cross offline, Rust implementation against the Java specification: pair **early-reply**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	1	32	3	violation	0.7	160	10 min, 16 GB	0.7	0.0	0.0	in-memory	brain01, 2026-11-02
2	4	512	12	violation	3.1	160	10 min, 16 GB	3.1	0.3	0.0	in-memory	brain01, 2026-11-02
3	36	18432	108	violation	3.9	158	10 min, 16 GB	3.9	1.8	0.0	in-memory	brain01, 2026-11-02

Not run (cross offline, Rust implementation against the Java specification). planned but never scheduled (grid points beyond the ladder): 5 early-reply bag, 5 correct bag, 6 early-reply bag, 6 correct bag, 7 correct bag.

cross online, Java implementation (JMC) against the Rust specification export

cross online, Java implementation (JMC) against the Rust specification export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	4	8	conforms	0.7	152	1 h, 250 GB		brain05, 2026-11-02
2	16	128	conforms	1.5	249	1 h, 250 GB		brain05, 2026-11-02
3	144	4608	conforms	33.9	325	1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	1	8	conforms	0.7	150	1 h, 250 GB		brain05, 2026-11-02
2	4	128	conforms	1.6	235	1 h, 250 GB		brain05, 2026-11-02
3	36	4608	conforms	37.7	1008	1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **early-reply**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1	4	32	24	violation	0.8	146	1 h, 250 GB	115	brain05, 2026-11-02
2	16	512	384	violation	3.4	285	1 h, 250 GB	161	brain05, 2026-11-02
3	144	18432	13824	violation	119.8	1296	1 h, 250 GB	224	brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **early-reply**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1	1	32	24	violation	0.9	163	1 h, 250 GB	130	brain05, 2026-11-02
2	4	512	384	violation	4.0	281	1 h, 250 GB	201	brain05, 2026-11-02
3	36	18432	13824	violation	149.4	418	1 h, 250 GB	241	brain05, 2026-11-02

Not run (cross online, Java implementation (JMC) against the Rust specification export). planned but never scheduled (grid points beyond the ladder): 1 early-reply fifo, 1 correct bag, 1 correct fifo, 1 early-reply bag, 2 correct bag, 2 early-reply bag, 2 early-reply fifo, 2 correct fifo, 3 correct bag, 3 early-reply bag, 3 correct fifo, 3 early-reply fifo, 5 correct bag, 5 early-reply bag, 6 correct bag, 6 early-reply bag, 7 correct bag.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	4	8	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01
2	16	128	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01
3	144	4608	conforms	0.6	4	1 h, 250 GB		brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	1	8	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01
2	4	128	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01
3	36	4608	conforms	0.7	4	1 h, 250 GB		brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair **early-reply**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1	4	32	24	violation	0.0	4	1 h, 250 GB	2	brain04, 2026-11-01
2	16	512	384	violation	0.1	2	1 h, 250 GB	2	brain04, 2026-11-01
3	144	18432	13824	violation	2.0	4	1 h, 250 GB	2	brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair **early-reply**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1	1	32	24	violation	0.0	4	1 h, 250 GB	2	brain04, 2026-11-01
2	4	512	384	violation	0.1	4	1 h, 250 GB	2	brain04, 2026-11-01
3	36	18432	13824	violation	2.1	4	1 h, 250 GB	2	brain04, 2026-11-01

Not run (cross online, Rust implementation (TraceForge) against the Java specification export). left out by the table rules (ok): 1 early-reply fifo, 1 correct bag, 1 correct fifo, 1 early-reply bag, 2 correct bag, 2 early-reply bag, 2 early-reply fifo, 2 correct fifo, 3 correct bag, 3 early-reply bag, 3 correct fifo, 3 early-reply fifo.

planned but never scheduled (grid points beyond the ladder): 5 early-reply bag, 5 correct bag, 6 early-reply bag, 6 correct bag, 7 correct bag.